



Arab Academy for Science, Technology, and Maritime Transport
College of Computing and Information Technology
Smart Village

Project Title

Prim vs Kruskal MST Playground & Huffman Coding Compression
Tool

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1. Overview:

This project consists of **two interactive visualisation tools** built using Python and Tkinter:

1. **Prim vs Kruskal MST Playground** – Allows users to visually compare Prim's and Kruskal's algorithms on randomly generated weighted graphs.
2. **Huffman Coding Compression Tool** – Demonstrates how text can be compressed using Huffman coding, visualising the Huffman Tree construction and encoding process.

2. Tech Stack and Requirements

- **Language:** Python 3.x
- **Libraries:**
 - ◊ networkx
 - ◊ matplotlib
 - ◊ tkinter (GUI)
 - ◊ threading, random, heapq, time

3. Huffman Coding Compression Tool

Description:

A GUI-based tool that compresses input text using Huffman coding and visualizes:

- Frequency analysis
- Huffman Tree structure
- Character codes
- Encoded and decoded strings
- Tree-building animation
- Character path tracing for each encoded value

Features:

- Real-time tree generation
- Step-by-step animation of tree merge operations
- Buttons to trace Huffman code for individual characters
- Tree visualization with labels and edges (0/1)

Example:

Frequencies: a:4, b:2, c:1, d:1

Codes: a: 0, b: 10, c: 110, d: 111

Encoded: 01001100100111

Decoded: abacabad

4. Prim vs Kruskal MST Playground

Description:

An interactive playground where users can:

- Generate a random weighted graph
- Run **Prim's** or **Kruskal's** algorithm step-by-step
- View a visual comparison of the paths and total weights selected by each algorithm

Features:

- Visual graph generation using `networkx.spring_layout`
- Animation of MST formation using:
 - Green edges for selected MST edges
 - Red edges for rejected ones
- Live update of:
 - Total MST weight
 - Priority queue for Prim
- Step-by-step edge evaluation with delay

Example Output:

Prim Total Path: 0-1-3-5-4-2

Total Weight: 21

Kruskal Total Path: 3-1-4-5-2-0

Total Weight: 21

5. UI Preview

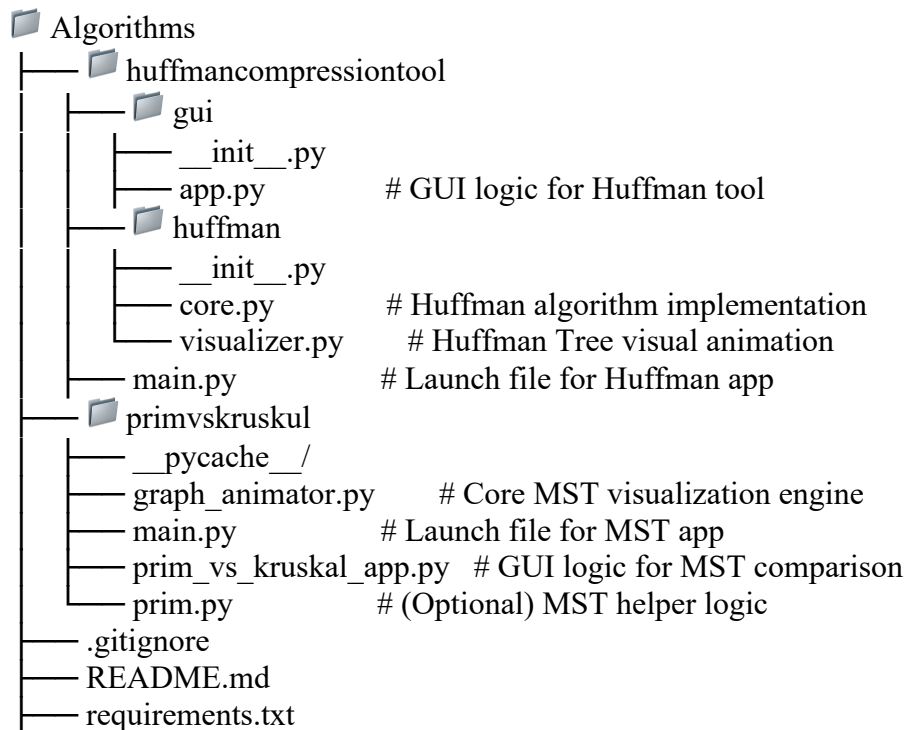
Prim vs Kruskal Playground

- Buttons: "Start Prim Animation", "Start Kruskal Animation", "Generate New Tree"
- Visual graph canvas with animated MST construction

Huffman Compression Tool

- Entry box for text input
- Visual Huffman Tree
- Code buttons for character path animation
- Frequencies, Codes, Encoded/Decoded view

6. Folder structure



7. GitHub Link:

<https://github.com/rolahadidy44/Algorithms>

8. Topics Covered:

- Greedy Algorithms

- Minimum Spanning Trees (Prim's and Kruskal's)
- Priority Queues
- Huffman Coding
- Tree Traversal and Visualization